

# Gustavo Fernandes Morais

Full Stack Developer · Itabira, MG, Brazil

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 Português |  English

## SUMMARY

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Computer Engineer graduated from the Federal University of Itajubá (UNIFEI), working across the full software development lifecycle: from front-ends in React, Angular and Razor to back-ends in .NET, Python, Node.js and PHP, covering relational and NoSQL data modelling, REST and WebSocket integrations, and delivery to production. Solid experience modernising legacy systems into current architectures and building serverless and generative AI applications on AWS, holding 3 AWS and 2 Google Cloud certifications. Keen to deepen my knowledge of distributed systems architecture, DevOps and Artificial Intelligence applications.

## PROFESSIONAL EXPERIENCE

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### Cloud Engineer & Solutions Architect, :upd8

Mar 2026 – Present

- Modernisation of legacy .NET applications to .NET Core 8 and 10 for enterprise clients: analysis of the existing source code, rewrite of controllers, Razor views and the data access layer, containerisation with Docker and release through Azure Pipelines.
- Modernisation of a PHP accounting platform (TYPO3 Flow 2.3 / Doctrine) to Laravel 11 with Eloquent and a DDD architecture, reorganising roughly 900k lines of legacy code into 20 domains and 381 models.
- Migration of SQL Server databases to PostgreSQL using AWS DMS and Schema Conversion, including the manual rebuild of foreign keys, triggers and stored procedures that were not converted automatically.
- Use of AWS Transform and Kiro to accelerate legacy code refactoring, with technical review of the generated output and remediation of security, routing and Linux compatibility regressions.
- Design and provisioning of AWS infrastructure following the Well-Architected Framework, from architecture diagrams through to production delivery.
- Weekly technical meetings with clients, technical documentation and architecture diagrams, plus support to the sales team in putting together technical proposals.

### Software Developer (Intern), :upd8

Apr 2025 – Mar 2026

- Development of full stack applications for clients and internal processes: front-ends in React 19 with TypeScript and Vite consuming REST APIs written in Python (FastAPI) and Node.js.
- Implementation of REST and WebSocket APIs on Amazon API Gateway, with Python Lambda functions, DynamoDB persistence and asynchronous processing through SQS.
- Design and delivery of AI solutions with Amazon Bedrock and, later, AWS AgentCore integrated with the Strands Agents framework, including RAG pipelines and agents with custom tools.
- Infrastructure as code automation with AWS CDK, Terraform and AWS SAM, along with CI/CD pipelines in GitHub Actions.
- Participation in requirements elicitation meetings covering functional and non-functional requirements, scope definition and architecture diagrams.
- Completion of an intensive cloud computing training programme, earning 5 professional certifications (3 AWS and 2 GCP).

### Web Development & IT Intern, ON Energia Solar

Sep 2023 – Feb 2024

- Development and maintenance of the company website using WordPress and PHP.
- Data collection and processing with Microsoft Power BI to provide insights to the team.
- Creation and management of Google Ads campaigns and support in rolling out internal IT solutions.

## Front-end Developer, Itabira City Hall

Jul 2022 – Jun 2023

- Development of the Itabira Tourism Portal with ASP.NET Razor, building the application interfaces with a focus on cross-platform access.
- Implementation of the core business logic in C# and of the interactions with a SQL Server database.
- Teamwork applying web design and project management practices within a public sector organisation.

## PROJECTS

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### AI Solutions Advisor

*React 19, TypeScript, FastAPI, Bedrock*

Full stack chat application backed by an AI agent. Front-end in React 19 with TypeScript and Vite, featuring the conversation interface and a debug panel for tool calls; back-end in Python with FastAPI and Pydantic validation, exposing the `/api/chat` and `/health` endpoints. The Strands agent running on Amazon Bedrock is equipped with custom tools for calculation, date/time and catalogue lookup. Tested with Vitest and Testing Library on the front-end and pytest with Hypothesis on the back-end.

### PDPoint system modernisation

*ASP.NET Core 8, C#, PostgreSQL, Docker*

Migration of an ASP.NET MVC application from .NET Framework to ASP.NET Core 8: over 20 controllers and 180 Razor views rewritten, data layer switched to Npgsql (PostgreSQL), structured logging with Serilog, object mapping with AutoMapper, PDF generation with PuppeteerSharp and containerisation for deployment through Azure Pipelines.

### Number Portability and Facade APIs

*Node.js, TypeScript, Express, Prisma*

Modernisation of legacy APIs for a telecommunications carrier. The Facade API was rewritten in Node.js 22 with TypeScript, Express and Prisma, organised in Clean Architecture (entities, use cases, handlers), with resilient integrations using retry and circuit breaker, Vitest test coverage, secrets and storage on Google Cloud and automated deployment. The Portability API was modernised in .NET with background jobs, webhooks and execution on Kubernetes.

### Multi-tenant conversational agent platform

*AWS CDK, Python, Bedrock*

WhatsApp chatbot with complete per-client isolation. Webhooks arrive through API Gateway and trigger an ACK Lambda that consolidates message fragments in a DynamoDB buffer and enqueues them in SQS with a delay; the invocation Lambda dynamically retrieves the client configuration and answers through Bedrock using a dedicated Knowledge Base. EventBridge keeps the vector store in sync, and every transaction records conversation tracing and token consumption for cost allocation. Infrastructure entirely in AWS CDK (Python), deployed via GitHub Actions.

### Intelligent document validation and extraction

*Python, Lambda, Textract, Terraform*

Serverless pipeline that validates authenticity and extracts structured fields from 12 groups of Brazilian documents, combining Rekognition, Textract and Bedrock. Modular architecture with dedicated validators and extractors per document type following the Template Method pattern, a REST API documented in OpenAPI, Lambda Layers for dependencies and infrastructure provisioned with Terraform.

### Personal projects

[github.com/GustavoF3rnandes](https://github.com/GustavoF3rnandes)

Notes-React (React + TypeScript), Food-ReactNative (React Native + TypeScript), a Fire Test Bench Simulator (C#), spreadsheet data processing in Python and a personal portfolio published on Vercel.

## EDUCATION

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2021 – 2026 B.Sc. in Computer Engineering at the **Federal University of Itajubá (UNIFEI)**

## CERTIFICATIONS

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AWS Certified Developer – Associate

AWS Certified Advanced Networking – Specialty

AWS Certified Cloud Practitioner

Google Professional Cloud Architect

Google Associate Cloud Engineer

## TECHNICAL SKILLS

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|----------------|---------------------------------------------------------------------------------------------------------------------------------------------|
| Front-end      | React, React Native, TypeScript, JavaScript, Angular, Razor / ASP.NET MVC, HTML5, CSS3, Bootstrap, Vite                                     |
| Back-end       | C# and .NET (Framework, 8 and 10), Python (FastAPI), Node.js (Express), PHP (Laravel, WordPress)                                            |
| Data           | SQL Server, PostgreSQL, DynamoDB, Prisma, Eloquent, Entity Framework, AWS DMS                                                               |
| Applied AI     | Amazon Bedrock, Strands Agents, AWS AgentCore, RAG and Knowledge Bases, Textract, Rekognition                                               |
| Cloud & DevOps | AWS, Google Cloud, Docker, Kubernetes, Terraform, AWS CDK, AWS SAM, GitHub Actions, Azure DevOps                                            |
| Practices      | REST and WebSocket APIs, OpenAPI, Clean Architecture, DDD, automated testing (pytest, Vitest, Testing Library), Git, requirements gathering |
| Languages      | Portuguese (native), English (professional working), Spanish (elementary)                                                                   |